

A Sharp Strong-Selection Obstruction to Universal Simultaneous Amplification

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Abstract

We ask whether a finite connected undirected weighted graph can strictly increase uniformly initialized single-mutant fixation probability, relative to the same-order complete graph, under both birth–death and death–birth updating for every mutant fitness $r > 1$. We prove a stronger negative result: no such graph is an all-fitness amplifier even for death–birth updating alone. If the positive-weight support is incomplete, its death–birth fixation probability has a strictly smaller strong-selection limit. If the support is complete, its first $1/r$ correction is no better than the complete graph’s, and the excess loss has an explicit sum-of-squares certificate. Equality forces all edge weights to be equal, in which case the process is exactly the complete-graph baseline. The obstruction applies to every finite graph and therefore rules out every fitness-independent graph family.

1 Relation to prior bounds

Tkadlec et al. proved that every noncomplete population structure—even with directed or weighted edges—is eventually dB-suppressing as fitness grows [1]. Their theorem explicitly leaves one residual possibility: a nonuniform weighted version of the complete graph. They list universal dB amplification as an open direction and observe that any example would have to lie in that class. The complete-support branch of our theorem closes this residual case for finite undirected symmetric loopless weights. Its $1/r$ coefficient also provides an equality characterization not supplied by the earlier average-degree bound.

Transient dB amplification is known and can be diagnosed at weak selection using coalescing random walks [2]. More recently, weighted modular graphs have been proved to amplify both Bd and dB selection on the fixed fitness interval $(1, 1.2)$ [3]. Those positive results are consistent with the obstruction here: our conclusion concerns a single graph required to amplify for every $r > 1$, and failure occurs in its strong-selection tail.

2 Model and result

Let $G = (V, w)$ be a connected loopless undirected weighted graph, with $|V| = n \geq 2$, $w_{ij} = w_{ji} \geq 0$, and every weighted degree positive. Mutants have fitness $r > 1$ and residents fitness one. Under death–birth updating a uniformly chosen vertex dies, after which a neighbor is chosen for reproduction with probability proportional to fitness times edge weight. Write $\rho_{\text{dB}}(G, r)$ for fixation probability averaged uniformly over all single-mutant initial vertices. Let

$$s_i = |\{j : w_{ij} > 0\}|$$

be the degree of i in the positive-weight support.

Theorem 1 (sharp strong-selection obstruction). *For every finite connected undirected weighted graph G on $n \geq 2$ vertices:*

1.

$$\lim_{r \rightarrow \infty} \rho_{\text{dB}}(G, r) = \frac{1}{n} \sum_i \frac{s_i}{s_i + 1}.$$

If the support is incomplete, then

$$\lim_{r \rightarrow \infty} (\rho_{\text{dB}}(K_n, r) - \rho_{\text{dB}}(G, r)) = \sum_i \frac{n-1-s_i}{n^2(s_i+1)} > 0.$$

2. *Suppose the support is complete and $n \geq 3$. Put*

$$\mathcal{E}(G) = \sum_i \sum_{\substack{j < k \\ j, k \neq i}} \frac{(w_{ij} - w_{ik})^2}{w_{ij} w_{ik}}.$$

Then

$$\rho_{\text{dB}}(K_n, r) - \rho_{\text{dB}}(G, r) = \frac{\mathcal{E}(G)}{n^2(n-2)r} + O(r^{-2}).$$

Moreover, $\mathcal{E}(G) = 0$ if and only if all off-diagonal weights are one common positive constant.

3. *For $n = 2$, every admissible graph is a global rescaling of K_2 , and $\rho_{\text{dB}}(G, r) = 1/2$ for all r .*

Consequently every graph not equivalent to K_n is strictly dB-suppressed for all sufficiently large finite r , while a scaled K_n ties the baseline for every r . No graph, and hence no fitness-independent graph family, is a strict dB amplifier for every $r > 1$; in particular universal simultaneous Bd/dB amplification is impossible.

The two branches of the certificate are shown schematically in Figure 1.

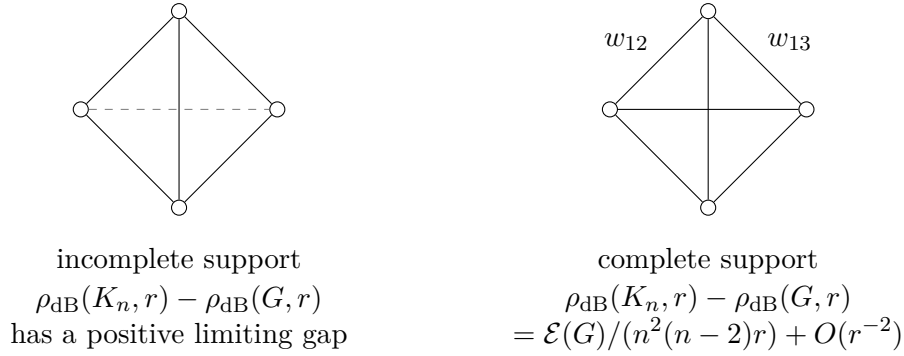


Figure 1: The exhaustive support dichotomy. The dashed segment on the left is a missing edge of the positive-weight support, not an edge weight to be optimized.

3 Complete-graph baselines

We derive both baselines directly. Let ϕ_k denote fixation probability on K_n from k mutants, and let p_k^+ and p_k^- be the probabilities of increasing and decreasing k . The first-step equations give

$$p_k^+(\phi_{k+1} - \phi_k) = p_k^-(\phi_k - \phi_{k-1}).$$

If $\gamma_k = p_k^-/p_k^+$, summing successive increments subject to $\phi_0 = 0$ and $\phi_n = 1$ gives

$$\phi_1 = \left(1 + \sum_{j=1}^{n-1} \prod_{k=1}^j \gamma_k \right)^{-1}.$$

For Bd updating, $\gamma_k = 1/r$, hence

$$\rho_{\text{Bd}}(K_n, r) = \frac{1 - r^{-1}}{1 - r^{-n}}. \quad (1)$$

For dB updating, direct counting gives

$$p_k^+ = \frac{n-k}{n} \frac{rk}{rk + n - k - 1}, \quad p_k^- = \frac{k}{n} \frac{n-k}{r(k-1) + n - k}.$$

Writing $A_k = n - 1 + (r - 1)k$, one has

$$\gamma_k = \frac{A_k}{rA_{k-1}}, \quad \prod_{k=1}^j \gamma_k = \frac{A_j}{(n-1)r^j}.$$

The finite sum telescopes to the exact formula

$$\rho_{\text{dB}}(K_n, r) = \frac{n-1}{n} \frac{1 - r^{-1}}{1 - r^{-(n-1)}}. \quad (2)$$

For $n \geq 3$ this has the strong-selection expansion

$$\rho_{\text{dB}}(K_n, r) = \frac{n-1}{n} - \frac{n-1}{nr} + O(r^{-2}). \quad (3)$$

We will use the following elementary finite-chain perturbation fact to make the $r \rightarrow \infty$ passage precise.

Lemma 2 (finite-state perturbation). *Let $P(\varepsilon)$ be a finite family of transition matrices continuous at $\varepsilon = 0$. Suppose that, from a specified initial state, all states reachable before absorption at $\varepsilon = 0$ lie in a set $\mathcal{R}$, the transient restriction $Q_{\mathcal{R}}(0)$ has spectral radius less than one, and the one-step probability of leaving $\mathcal{R}$ is $O(\varepsilon)$. Then the absorption probabilities converge to those of $P(0)$. If $\mathcal{R}$ is the full transient state space and the entries are analytic, then the absorption probabilities are analytic near zero.*

Proof. The inverse $(I - Q_{\mathcal{R}}(\varepsilon))^{-1}$ exists and remains bounded for all sufficiently small ε . Multiplying it by the vector of one-step leakage probabilities shows that the probability of leaving $\mathcal{R}$ before absorption is $O(\varepsilon)$. The absorption equations within $\mathcal{R}$ therefore converge to the nonsingular limiting system; behavior after leakage can change a hitting probability by at most the leakage probability. In the full-space analytic case, $(I - Q(\varepsilon))^{-1}$ is analytic by the adjugate formula. $\square$

4 Incomplete positive-weight support

Fix an initial mutant vertex i . At $r = \infty$, death of i causes extinction. Death of any one of its s_i support-neighbors is followed by mutant reproduction with probability one and creates an adjacent mutant pair. Deaths at nonneighbors are self-loops. Conditioning on the first state change therefore gives probability $s_i/(s_i + 1)$ of reaching a pair before extinction.

Once an adjacent pair has formed, the mutant set remains support-connected. Every mutant has a mutant neighbor and hence cannot be replaced by a resident in the limiting chain. Whenever a boundary resident dies it becomes mutant. Connectedness of the support implies monotone growth to fixation almost surely. Thus

$$\lim_{r \rightarrow \infty} \rho_{\text{dB}}(G, r) = \frac{1}{n} \sum_i \frac{s_i}{s_i + 1}.$$

Subtracting this from $(n - 1)/n$ gives the exact termwise certificate below.

$$\frac{n - 1}{n} - \frac{1}{n} \sum_i \frac{s_i}{s_i + 1} = \sum_i \frac{n - 1 - s_i}{n^2(s_i + 1)}. \quad (4)$$

Every denominator is positive. If an edge is missing, at least one numerator is positive, proving the first branch of Theorem 1.

For completeness, the limiting argument is stable: from a singleton, every limiting state reached before absorption is a connected mutant set. The finite limiting transient chain absorbs almost surely, so its fundamental matrix is nonsingular. For finite $r = 1/\varepsilon$, a transition that leaks from this set requires a resident to replace a mutant having a mutant neighbor, and has probability $O(\varepsilon)$ in each fixed state. Lemma 2 therefore proves convergence of the finite- r hitting probabilities, including the effect of all paths after such leakage.

5 Complete support and the sum-of-squares correction

Assume $n \geq 3$ and $w_{ij} > 0$ for all $i \neq j$. Put

$$d_i = \sum_{j \neq i} w_{ij}, \quad \varepsilon = r^{-1}, \quad a_{ij} = \frac{d_i - w_{ij}}{w_{ij}}.$$

Let $q_S(\varepsilon)$ be extinction probability from mutant set S . At $\varepsilon = 0$, every set with at least two mutants fixates, while $q_{\{i\}}(0) = 1/n$. The limiting transient chain absorbs almost surely, so Lemma 2 shows that the hitting probabilities are analytic near $\varepsilon = 0$.

The first-order equations for $|S| \geq 3$ have no forcing: a rare loss still leaves at least two mutants, whose zeroth-order extinction probability is zero. Backward induction from the full state gives

$$q_S(\varepsilon) = O(\varepsilon^2), \quad |S| \geq 3.$$

For $S = \{i, j\}$, loss of mutant i occurs with probability

$$\frac{1}{n} \frac{\varepsilon(d_i - w_{ij})}{w_{ij} + \varepsilon(d_i - w_{ij})} = \frac{\varepsilon a_{ij}}{n} + O(\varepsilon^2),$$

and similarly for j . The leading self-loop probability is $2/n$ and the leading total gain probability is $(n-2)/n$. Since a resulting singleton has extinction probability $1/n + O(\varepsilon)$, the exact first-step equation gives, on putting

$$\begin{aligned} b_{ij} &= \frac{a_{ij} + a_{ji}}{n(n-2)}, \\ q_{\{i,j\}}(\varepsilon) &= b_{ij}\varepsilon + O(\varepsilon^2). \end{aligned} \tag{5}$$

From singleton $\{i\}$, death of resident j creates $\{i, j\}$ with probability

$$t_{ij} = \frac{1}{n} \frac{w_{ij}}{w_{ij} + \varepsilon(d_j - w_{ij})} = \frac{1}{n}(1 - a_{ji}\varepsilon) + O(\varepsilon^2).$$

Death of i causes extinction with probability exactly $1/n$; every other outcome is a self-loop. Hence

$$\left(\frac{1}{n} + \sum_{j \neq i} t_{ij} \right) q_{\{i\}} = \frac{1}{n} + \sum_{j \neq i} t_{ij} q_{\{i,j\}}.$$

Substitution of (5), followed by uniform averaging, yields

$$\begin{aligned} T(G) &= \sum_i \sum_{j \neq i} a_{ij}, \quad A(G) = \frac{T(G)}{n^2(n-2)}, \\ \rho_{\text{dB}}(G, r) &= \frac{n-1}{n} - \frac{A(G)}{r} + O(r^{-2}). \end{aligned} \tag{6}$$

For the unit complete graph, $a_{ij} = n-2$ and $T(K_n) = n(n-1)(n-2)$. The difference admits the exact identity

$$\begin{aligned} T(G) - n(n-1)(n-2) &= \sum_i \left[d_i \sum_{j \neq i} \frac{1}{w_{ij}} - (n-1)^2 \right] \\ &= \sum_i \sum_{\substack{j < k \\ j, k \neq i}} \frac{(w_{ij} - w_{ik})^2}{w_{ij}w_{ik}} = \mathcal{E}(G). \end{aligned} \tag{7}$$

All denominators in (7) are positive. Equality requires every star of incident weights to be constant; symmetry and complete support then force one global edge weight. Equations (3), (6), and (7) prove the second branch of Theorem 1.

6 Exact verification

The accompanying verifier is separate from the discovery implementation. It constructs all subset-state transitions directly over $\mathbb{Q}(r)$, checks row normalization and termwise positivity after the substitution $r = 1+x$, verifies strong lumpability of K_n by mutant count, solves the exact absorbing equations, and extracts the limiting coefficients.

As one nontrivial replay, take a weighted triangle with edge weights $(w_{12}, w_{13}, w_{23}) = (2, 3, 5)$. Exact elimination gives

$$\rho_{\text{dB}}(G, r) - \rho_{\text{dB}}(K_3, r) = -\frac{r(r-1)P(r)}{9(r+1)Q(r)},$$

where

$$\begin{aligned} P(r) &= 900r^4 + 5491r^3 + 9290r^2 + 5491r + 900, \\ Q(r) &= 675r^6 + 6600r^5 + 21791r^4 + 31768r^3 \\ &\quad + 21791r^2 + 6600r + 675. \end{aligned}$$

Every displayed denominator factor is positive and the numerator is strictly negative for $r > 1$. The verifier also recovers the strong coefficient $22/27 > 2/3$, and recovers $343/320 > 3/4$ for the complete four-vertex weighting with lexicographic edge weights $(1, 2, 3, 4, 5, 6)$. For incomplete support it recovers limits $7/12$ for the four-vertex path and $9/16$ for the four-vertex star, against the baseline $3/4$.

These calculations are checks, not evidence used to extend a finite search: the universal quantifier is discharged by the exact certificates (4) and (7).

7 Finite versus asymptotic statements

The expansions in Theorem 1 are taken separately for each fixed finite graph. They imply that for every nonbaseline G there exists a finite threshold R_G such that

$$\rho_{\text{dB}}(G, r) < \rho_{\text{dB}}(K_n, r) \quad (r > R_G).$$

No claim of a graph-independent threshold is needed. The all- r amplifier property quantifies over every finite $r > 1$ for each graph, so the existence of one such suppressing tail already rules it out. The equality case is exact, not asymptotic: global rescaling cancels from every neighbor-choice probability, so a uniformly weighted complete graph has precisely the baseline chain for all finite r .

References

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